Measures of Central Tendency

2026-04-17

Introduction

A measure of central tendency is a single number that summarises "where the data are". The three classical choices – arithmetic mean, median, and mode – are not interchangeable: each answers a slightly different question, and each is appropriate in a different context. Choosing the right one matters because readers of your results will form mental pictures of the data that the number implies.

Prerequisites

The reader should know the difference between a numeric, an ordered categorical, and a nominal categorical variable, and should be able to read a histogram. Familiarity with R vectors is assumed.

Theory

The **arithmetic mean** of observations $x_1, \ldots, x_n$ is $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$. It is the centre of mass of the data and minimises the sum of squared deviations from any candidate centre. The mean uses all observations, but it is sensitive to outliers: a single extreme value can drag it arbitrarily far from the bulk of the data.

The **median** is the value that separates the lower half of the data from the upper half: $x_{((n+1)/2)}$ for odd n, the average of the two middle values for even n. The median minimises the sum of absolute deviations. Unlike the mean, it is robust to outliers: moving the largest observation to infinity leaves the median unchanged.

The **mode** is the most frequently occurring value. For continuous data, the mode is typically defined from a kernel density estimate rather than from the raw values, because ties are unlikely. The mode is the only measure of central tendency that makes sense for nominal categorical variables.

The **trimmed mean** computes the mean after removing a proportion α of the smallest and largest observations – a compromise between the efficiency of the mean and the robustness of the median. A 10% trimmed mean removes the bottom and top 10% of the observations; a 50% trimmed mean is the median.

Assumptions

These are descriptive statistics, not tests, so they do not carry distributional assumptions. They do carry interpretive assumptions: the mean is most useful when the distribution is roughly symmetric; the median is preferred for skewed distributions and for ordinal data; the mode is preferred for categorical data and for multimodal distributions where a single centre is misleading.

R Implementation

```r
library(DescTools)
library(psych)

x <- c(2.1, 2.4, 2.5, 2.7, 2.9, 3.1, 3.3, 3.4, 3.7, 42.0)

mean(x)
median(x)
mean(x, trim = 0.1)
```

```r
Mode(x)
psych::geometric.mean(x)
psych::harmonic.mean(x)
```

The vector `x` is a small sample with a single outlier at 42. Running the above shows a mean of 6.71, a median of 3.00, and a 10% trimmed mean of 2.94. The geometric mean (3.44) and harmonic mean (3.15) are close to the median rather than the mean, since both give less weight to the outlier.

Output & Results

For a skewed or contaminated distribution, the three classical measures of central tendency can differ substantially:

Statistic	Value
Mean	6.71
Median	3.00
10% trimmed mean	2.94
Geometric mean	3.44
Harmonic mean	3.15

The mean is pulled far from the bulk of the data by the single outlier; every other measure stays close to the main cluster.

Interpretation

For a manuscript, the choice of statistic should be justified in the methods section. A paper reporting the "mean serum creatinine" in a cohort where several dialysis patients are included should either exclude those patients explicitly or report the median with IQR. Never mix conventions across papers in a series: if you reported medians in one table you should not swap to means in the next.

Practical Tips

- Plot the data before choosing the statistic. A histogram or boxplot will show immediately whether the mean is a sensible summary.
- Report the median and IQR (not the mean and SD) for variables with strong right skew: lab values, income, incubation times.
- For small samples ($n < 10$), any central tendency measure is uncertain; consider bootstrapping to get a confidence interval for the statistic of choice.
- Do not report the mode for continuous data unless it comes from a density estimate – the sample mode of continuous data is the lowest value in the tied set, which is meaningless.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1